

STMotif: An R Package for Spatial-Time Motif Discovery

Heraldo Borges¹, Antonio Castro¹, Rafaelli Coutinho¹, and Eduardo Ogasawara¹

¹ Federal Center for Technological Education of Rio de Janeiro (CEFET/RJ), Brazil

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Summary

STMotif is an R package for discovering, ranking, and visualizing frequent patterns, called motifs, in **spatial-time series**. Unlike traditional motif discovery tools that analyze each time series independently, STMotif targets patterns that are constrained simultaneously in space and time and may emerge only when neighboring series are analyzed jointly. The package implements the **Combined Series Approach (CSA)** introduced by Borges et al. (2020) and provides end-to-end support for normalization, SAX encoding, constrained motif search, multi-criteria ranking, and visual inspection of the resulting motifs, all accessible through a small set of composable R functions.

Statement of need

Many real-world phenomena can be represented as time series, and some of them are naturally associated with spatial position, producing spatial-time series datasets (Atkinson, 2000; Shumway & Stoffer, 2017). Examples include sensor networks, environmental monitoring, ocean currents, seismic data, and other geo-referenced temporal observations (Fu, 2011). In these scenarios, relevant patterns may not be strongly expressed in a single series, but rather distributed across multiple neighboring series.

A motif is a subsequence that appears frequently in time series (Torkamani & Lohweg, 2017; ?). Motif discovery aims to identify previously unknown and recurrent subsequences in time series (Jessica Lin et al., 2002; Mueen, 2014). Existing research has produced important advances in motif discovery algorithms and representations (Torkamani & Lohweg, 2017; Yeh et al., 2018), but these methods mainly focus on single-series analysis. This creates a gap for applications in which the pattern of interest is both temporally localized and spatially distributed.

STMotif addresses this gap by providing a software implementation tailored to **motifs constrained in space and time**. The package is intended for researchers and practitioners who need to identify recurrent spatiotemporal patterns, rank them according to informative criteria, and inspect them visually in the original data. The main audience includes users working with georeferenced temporal data such as sensor networks, environmental monitoring, and related spatial analytics applications.

State of the field

Traditional motif discovery methods generally assume that the input is a single series or a collection of series analyzed separately, analyzing only the temporal dimension. The CSA method proposed by Borges et al. (2020) extends this perspective by partitioning a spatial-time

dataset into spatial and temporal blocks, combining the observations within each block, and then searching for motifs that satisfy both occurrence and spatial-distribution constraints.

STMotif operationalizes this method in software and, to our knowledge, is the first R package focused on motif discovery under joint spatial and temporal constraints. This is therefore not a repackaging of an existing workflow: the contribution is the software realization of a distinct spatial-time motif mining method together with its ranking and visualization pipeline.

Software design

STMotif is organized as a pipeline that separates symbolic preprocessing, constrained motif search, ranking, and visualization, as illustrated in Figure 1. Each stage can be invoked independently, so that intermediate outputs — such as the SAX-encoded dataset or the unranked motif list — remain accessible for inspection or reuse.

Normalization and SAX encoding. Given a spatial-time dataset in which each time series is associated with a spatial position, direct motif discovery over real-valued subsequences is inefficient. The package therefore first normalizes each series using **z-score normalization** and then discretizes it via **SAX indexing** (Keogh & Ratanamahatana, 2005; J. Lin et al., 2007), producing symbolic words drawn from an alphabet of user-defined size a . This widely adopted representation enables efficient indexing and comparison of subsequences while preserving shape information relevant for motif discovery.

Constrained motif search. The indexed dataset is then partitioned into blocks according to two user-defined constraints: spatial block size (sb), the number of neighboring series per block, and temporal block size (tb), the length of the temporal window. Within each block, subsequences are combined into a single series and examined with a sliding window of size w . A candidate is accepted as a spatial-time motif only if it meets both a minimum occurrence count (σ) and a minimum count of distinct spatial series with occurrences (κ) inside the block (Borges et al., 2020).

Ranking. Motif mining typically returns many candidates. STMotif ranks them using three complementary criteria: number of occurrences, spatial-temporal proximity of occurrences (measured via a minimum spanning tree over pairwise distances), and Shannon entropy of the motif's symbolic content. The ranking stage helps users focus on the most informative patterns without losing access to the complete result set.

Visualization. Two plotting functions help users connect symbolic motifs back to the original data: one highlights motif occurrences on individual series, the other displays a heat map of the full dataset with motif positions color-coded.

Figure 1: Main functionalities of STMotif, including normalization and SAX indexing, motif search, ranking, and visualization.

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The package exposes this workflow through a compact set of user-facing functions:

- `NormSAX()` applies normalization and SAX encoding to the input dataset.
- `SearchSTMotifs()` discovers motifs from the original and encoded datasets using the parameters w , a , sb , tb , σ , and κ .
- `RankSTMotifs()` ranks the discovered motifs according to occurrence count, proximity, and entropy.
- `CSAMiningProcess()` executes the complete workflow in a single call.
- `display_motifsSTSeries()` visualizes motifs over selected spatial-time series.
- `display_motifsDataset()` displays motif occurrences over the whole dataset.

80 This design separates preprocessing, mining, ranking, and visualization, while also allowing the
81 full pipeline to be executed compactly when desired. In practice, this supports both exploratory
82 use by domain researchers and scripted, reproducible use in larger analyses.

83 Research impact statement

84 STMotif operationalizes the CSA method published by Borges et al. (2020) as reusable
85 research software, lowering the barrier for motif discovery in spatial-time datasets. Beyond
86 motif identification, the package emphasizes interpretability through dedicated visualization
87 functions, which help users inspect the spatial-temporal distribution of the discovered patterns
88 and connect symbolic motifs back to the original data.

89 This capability is relevant for scientific and applied domains in which analysts need to detect
90 recurring local phenomena that are not visible when each series is processed in isolation. By
91 packaging discovery, ranking, and visualization into a single R tool distributed through CRAN,
92 STMotif supports reproducible experimentation, method reuse, and practical adoption of
93 spatial-time motif mining. The associated CSA publication provides direct evidence of research
94 use and scientific relevance.

95 Example usage

96 The package ships with a synthetic dataset of 36 spatial-time series, each containing 40
97 observations. The following example runs the complete CSA pipeline and visualizes the results:

```
library(STMotif)
D <- STMotif::example_dataset # loads D (40 x 36 array)

# Step 1: Normalize and apply SAX indexing (alphabet size = 5)
DS <- NormSAX(D, a = 5)

# Step 2: Search for spatial-time motifs
stmotifs <- SearchSTMotifs(D, DS, w = 4, a = 5, sb = 4, tb = 10, si = 2, ka = 2)

# Step 3: Rank the discovered motifs
rstmotifs <- RankSTMotifs(stmotifs)

# Step 4: Visualize motifs on series ST1 through ST9
display_motifsSTSeries(D, rstmotifs, space = c(1:9))

# Alternative: heat map of the full dataset
display_motifsDataset(D, rstmotifs, a = 5)
```

98 The search identifies six instances of “ceeb”, seven instances of “bbed”, and four instances
99 of “baba”—distributed across the spatial-time blocks. The ranking prioritizes motifs that are
100 frequent, spatially and temporally concentrated, and information-rich. The entire pipeline can
101 also be executed compactly, in a single call:

```
rstmotifs <- CSAMiningProcess(D, w = 4, a = 5, sb = 4, tb = 10, si = 2, ka = 2)
```

102 Visualization reveals that some motifs span a broad spatial range at a fixed time (temporally
103 constrained), while others are spatially restricted. STMotif provides two complementary
104 visualization functions. The first, `display_motifsSTSeries()`, overlays the discovered motifs
105 on the original time series, highlighting each motif group with a distinct color (Figure 2):

```
display_motifsSTSeries(D, rstmotifs, space = c(1:9))
```

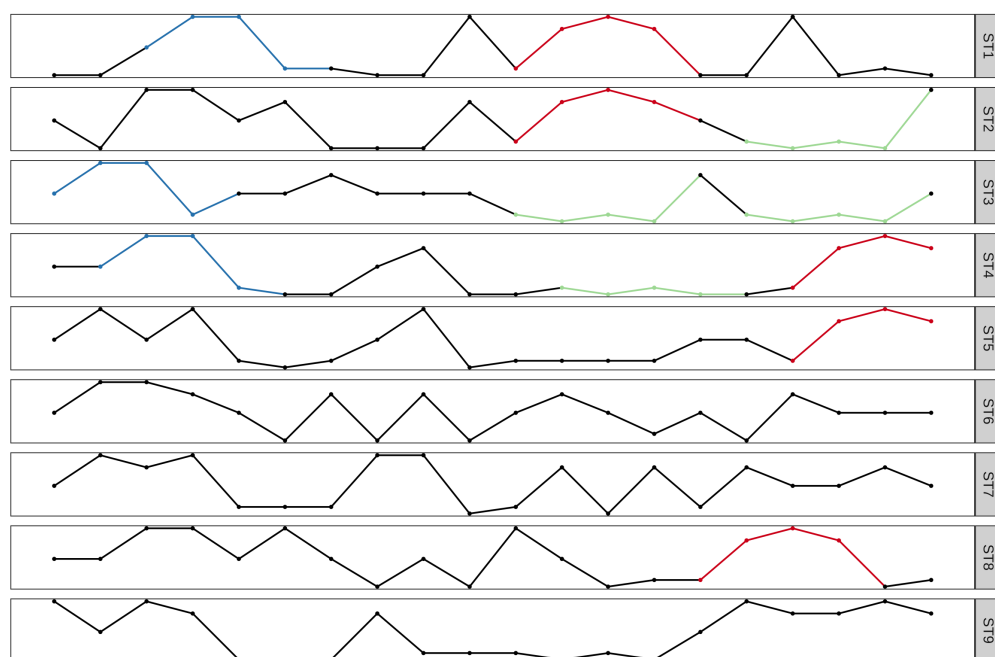


Figure 2: Spatial-time series ST1 through ST9 with ranked motif groups highlighted. Each color corresponds to a different motif group. Because the mining operates on SAX-encoded data, motif instances sharing the same symbolic word may differ in their real-valued shape.

106 The second function, `display_motifsDataset()`, produces a heat map of the SAX-encoded
107 dataset with motif occurrences color-coded over their spatial and temporal positions (Figure 3):

```
display_motifsDataset(D, rstmotifs, a = 5)
```

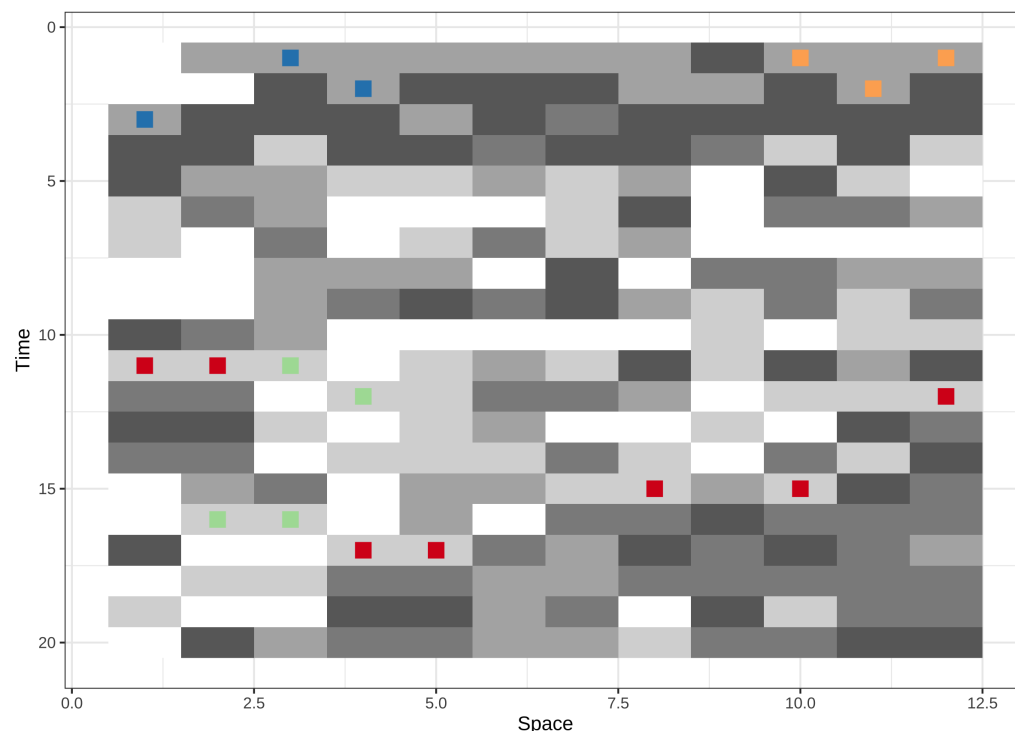


Figure 3: Heat map of the full spatial-time series dataset. Grayscale intensity represents the SAX-encoded values; colored regions mark the spatial-temporal positions of each motif group. Motifs in red (“ceeb”) span a broad spatial range at a fixed time, while motifs in blue (“baba”) are spatially restricted. These spatially distributed patterns would not be detected by traditional methods that analyze each series independently.

Availability and community guidelines

STMotif is available on CRAN and can be installed with `install.packages("STMotif")`. The package is distributed under the GPL-3 license. Its official documentation, including the reference manual, is accessible via the CRAN project page, available at <https://cran.r-project.org/package=STMotif>. Users can report bugs and request new features through the channels indicated on the CRAN page, where up-to-date information about the package and its maintenance is also provided.

AI usage disclosure

Generative AI tools were used only for language support during manuscript adaptation to the JOSS format. The authors reviewed and verified the resulting text, and the technical content, software description, methodological claims, and conclusions remain the responsibility of the authors.

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